

Internet Computing

Introduction to HTML Canvas

Gibson Lam

HTML Canvas

- HTML canvas is a drawing area inside a web page, which is created by the `<canvas>...</canvas>` element
- You draw things inside it using JavaScript
- It handles bitmap content only, i.e. you can see its 'pixels' if you zoom in to the area
- It is especially good for making complex visualizations and games in web pages

Canvas or SVG

- Basic reason – you choose canvas if you want bitmap content, whereas you choose SVG if you want vector graphics
- Other reasons to choose canvas over SVG:
 - You want to create highly dynamic content
 - You handle a lot of moving things
 - You want to create 2D and/or 3D content

The Canvas Element

- The `<canvas>...</canvas>` element is a rectangular area that can be created like this:

```
<canvas width="600" height="400">  
</canvas>
```

- The above HTML creates a 600 by 400 empty rectangular area in the web page
- You add content inside the area using JavaScript later

Using JavaScript

- For example, you can create your canvas content in the jQuery ready event:

...

```
<canvas width="600" height="400"></canvas>
```

...

```
<script>
```

```
$(function() {  
    let cv = $("canvas").get(0);  
    ...creating the canvas content...
```

```
});
```

```
</script>
```

...



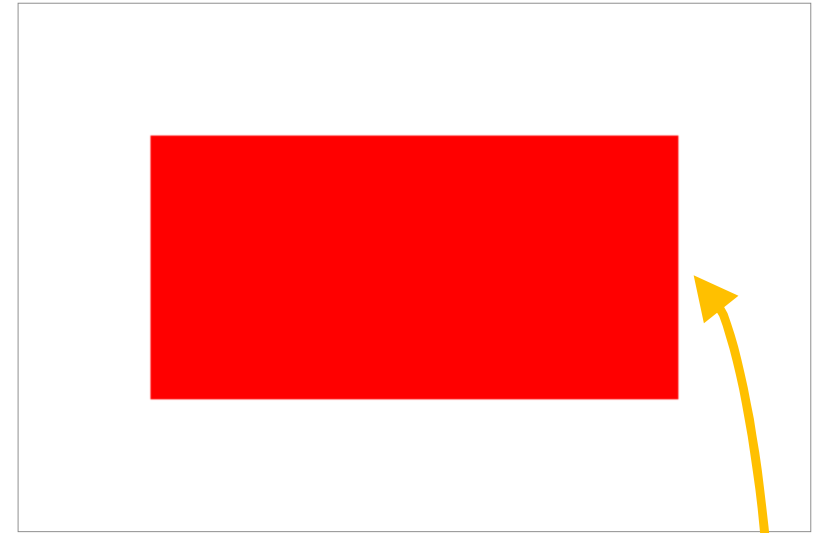
*Get the canvas
element*

Getting the Context

- You need to get a 'drawing context' before starting to draw things in canvas
- There are two types of contexts:
 - Getting a 2D context for 2D content
`cv.getContext("2d");`
 - Getting a WebGL context for 3D content
`cv.getContext("webgl");`
- In our course, we will only look at how to use canvas to create 2D content

A 2D Example

- Here is an example that gives you a red rectangle inside a canvas area:



```
<canvas width="600" height="400"></canvas>
```

```
...
```

```
<script>
```

```
$(function() {
```

```
    let cv = $("canvas").get(0);
```

```
    let context = cv.getContext("2d");
```

```
    context.fillStyle = "red";
```

```
    context.fillRect(100, 100, 400, 200);
```

```
});
```

```
</script>
```

*Drawing a
red rectangle*

Using the 3D Context

- Using the 3D context, you can create and manipulate 3D objects inside canvas
- It uses WebGL, which is an API based on the OpenGL 3D API
- Working with 3D visualization involves additional programming languages so we won't go into the details in this course
- An example is shown on the next slide

An Example 3D Cube

- You can rotate, move or zoom the cube by a combination of the mouse button, the Shift key and the Ctrl key

